

Parathyroid Disorders I: Hyperparathyroidism & Hypercalcaemia

Internal Medicine Department | Source: 7-Hyperparathyroidism.pdf (40 slides)

Legend: ★ = high-yield · [added] = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 Bridge from earlier lectures:

- **Hormones & receptors / regulatory systems (Physiology):** PTH is a **peptide** hormone in a classic negative-feedback loop; Ca^{2+} is the controlled variable.
- **Thyroid 2: medullary thyroid cancer** belongs to **MEN 2A** together with **parathyroid hyperplasia**. Thyroid surgery is the commonest cause of the opposite problem, **hypoparathyroidism** (next lecture).
- **Lung cancer: squamous cell carcinoma** → **PTHrP** → **hypercalcaemia** (the paraneoplastic counterpart of this lecture).
- **Hypertension:** primary hyperparathyroidism was on the endocrine HTN list, and **thiazides raise Ca** while loops lower it.
- **Gastric tumours / PUD:** peptic ulcer + hypercalcaemia → think **MEN 1 / Zollinger-Ellison**.

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Pathophysiology and etiology of parathyroid disorders	1ry vs 2ry vs 3ry, ectopic
Knowledge	★ Epidemiology, manifestations, complications and management of hyperparathyroidism	"Stones, bones, groans, moans"; surgery indications
Knowledge	★ Calcium homeostasis	PTH / vit D / calcitonin table; corrected Ca
Knowledge	★ Causes and manifestations of hypercalcaemia	Differential MCQ
Skill	Interpret clinical findings and investigations → diagnosis	Lab pattern (Ca, PO_4 , ALP, PTH)
Skill	Evidence-based management plan for HPT	Surgery vs medical
Skill	★ Recognise emergencies (hypercalcaemic crisis)	Saline first, then furosemide

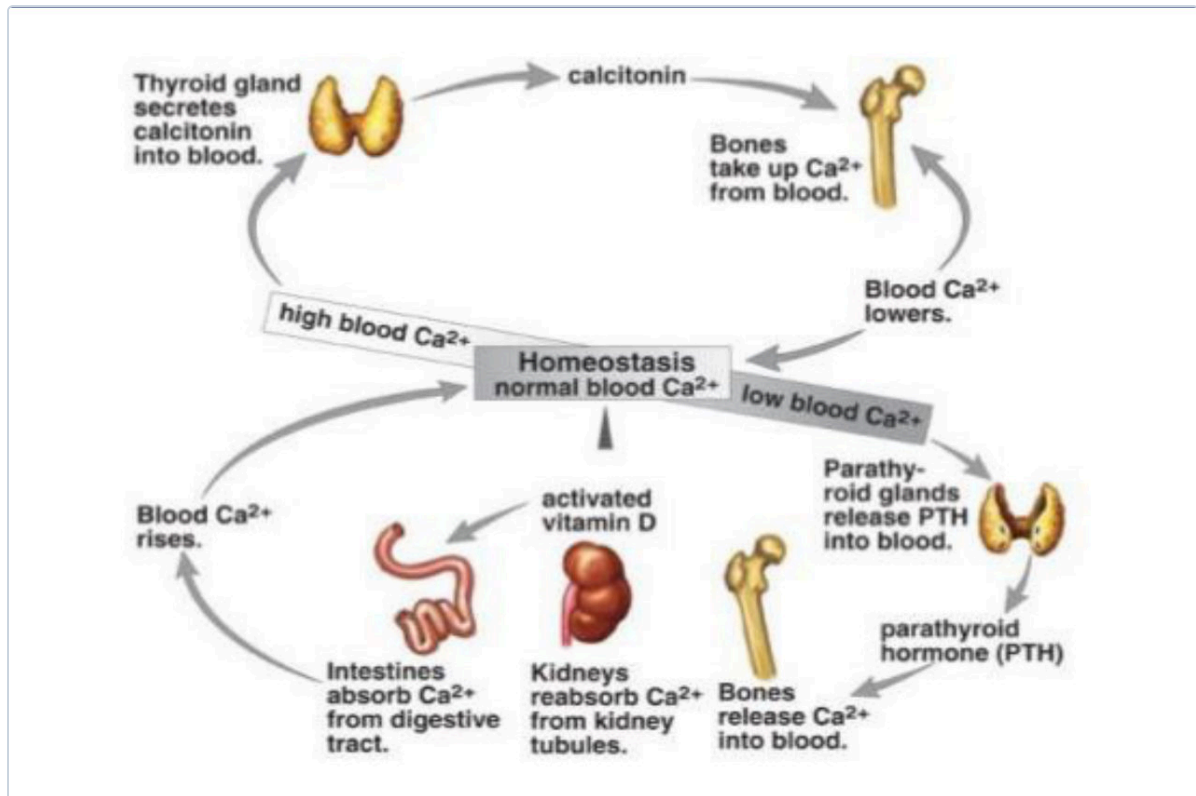
Lecture flow: Calcium homeostasis (PTH, vit D, calcitonin; assessment; functions) → Hyperparathyroidism (causes, clinical picture, osteitis fibrosa cystica, investigations, radiology, localisation, Dent test) → Treatment (surgery, hungry bone, medical) → Hypercalcaemic crisis → Differential diagnosis (causes of hypercalcaemia; generalised bone disease) → Case → Summary.

2. Main Content (Lecture Order)

2.1 Calcium Homeostasis (Slides 4-6) ★

- **Low blood Ca^{2+}** → parathyroid glands release **PTH** → **bones** release Ca^{2+} , **kidneys** reabsorb Ca^{2+} and **activate vitamin D**, **intestine** absorbs Ca^{2+} → blood Ca^{2+} rises.

- **High blood Ca^{2+}** → thyroid **C cells** secrete **calcitonin** → bones take up Ca^{2+} → blood Ca^{2+} falls.
- Serum Ca and PO_4 are controlled by three hormones: **parathormone, calcitonin, active vitamin D**.
- Regulation involves three sites: **bone, intestine, kidney**.



Slide 5: physiological background of calcium regulation

Hormone actions (slide 6):

Site	PTH*	Vitamin D	Calcitonin
Absorption (intestine)	↑ Ca, ↓ P	↑ Ca, ↑ P	↓ Ca
Reabsorption (kidney)	↑ Ca, ↓ P (phosphaturia)	↑ Ca, ↑ P	↓ Ca, ↓ P
Resorption (bone)	↑ Ca, ↑ P	↑ Ca, ↑ P	↓ Ca, ↓ P
★ Blood (net result)	↑ Ca, ↓ P	↑ Ca, ↑ P	↓ Ca, ↓ P

* **PTH acts directly on bone and kidney, and indirectly on the intestine** (through activation of vitamin D: 1α -hydroxylase in the kidney [added]).

	Parathormone*		vitamin D		Calcitonin	
Absorption (intestine)	↑ Ca	↓ P	↑ Ca	↑ P	↓ Ca	
Reabsorption (kidney)	↑ Ca	↓ P	↑ Ca	↑ P	↓ Ca	↓ P
Resorption (bone)	↑ Ca	↑ P	↑ Ca	↑ P	↓ Ca	↓ P
Blood (net result)	↑ Ca	↓ P	↑ Ca	↑ P	↓ Ca	↓ P

*PTH acts directly on bone & kidney & indirectly on intestine (through activation of vit.D).

Slide 6: PTH, vitamin D and calcitonin effects

Mnemonic: PTH = "Phosphate Trashing Hormone" [added] → raises Ca but **dumps PO₄ in the urine**. Vitamin D raises **both**. Calcitonin lowers **both**.

2.2 Assessment of Calcium (Slides 7-8) ★

Fraction	% of serum Ca	Note
Body distribution	99% in bone, 1% extracellular	
Ionized (active)	50%	The physiologically active part
Non-ionized (reserve): bound to albumin	40%	Changes with albumin
Non-ionized: other complexes	10%	Carbonate, phosphate...

- **Normal total Ca: 8.4-10.2 mg%** (slide 7) / **8.5-10.5 mg/dL** (slides 7-8).
- **Ionized Ca normal: 4.65-5.25 mg/dL** (slide 8). The slide labels **< 4.5 mg/dL** as low.
- ★ **Liver cirrhosis and nephrotic syndrome** → ↓ albumin → **low total Ca but NO tetany**, because **ionized Ca is still normal**.
- Ca is closely related to **P** (normal **3-4.5 mg%**). The slide states **Ca × P = constant (~40)**.

★ **Corrected Ca = serum Ca + 0.8 × (normal albumin – patient's albumin)**. Add **0.8 mg/dL of Ca for each 1 g/dL of albumin below 4** (the slide writes "mg/dl" for albumin; it means g/dL).

[added] Worked example: Ca 7.6, albumin 2.0 → 7.6 + 0.8 × 2 = 9.2 (normal). No treatment needed.

- Normal Ca level : 8.4 – 10.2 mg%.

A. Ionized (active) : 50%

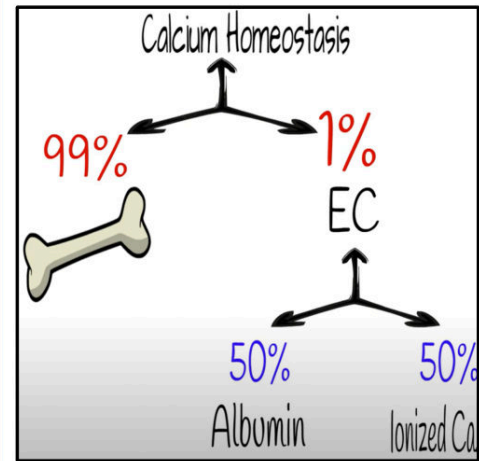
B. Non-ionized (reserve) : 50%

- 40% bound to albumin. - 10% any other bond (Ca carbonate, phosphate)

Liver cirrhosis & nephrotic syndrome → ↓ albumin → hypocalcaemia ,
but no tetany because ionized Ca is still normal.

- Regulation of Ca :

- Ca level is closely affected & related to P [n. 3 - 4.5 mg%]
- Notice that $Ca \times P = \text{constant}$ [40].



- Total serum Calcium level** (8.5-10.5 mg/dL) after correction = (Correction of serum Ca for hypoalbuminemia **Corrected by the level of albumin**).

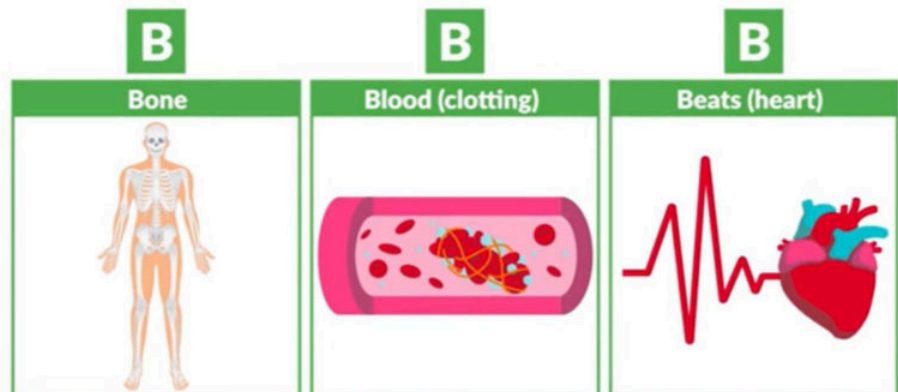
1- By adding 0.8 mg/dl of Ca for each 1 mg/dl of albumin reduction below 4 mg/dl.

Corrected Ca = serum Ca + 0.8 x (Normal albumin - Pts' albumin)

Slide 7: calcium fractions and correction for albumin

Functions of calcium (slide 8). Mnemonic "C keeps the 3 B's strong":

- C** = Contracts the muscles
- B**one
- B**lood (clotting, factor IV)
- B**eats (heart)



Keeps the 3 B's Strong

- B** - Bone
- B** - Blood (clotting)
- B** - Beats (heart)

Total serum Calcium level (8.5-10.5 mg/dL)

Serum ionized Ca < 4.5 mg/dL (Normal: 4.65-5.25 mg/

Slide 8: functions of calcium (3 B's)

🔄 **Recap: Homeostasis.** PTH: ↑ Ca, ↓ PO₄ (direct on bone and kidney, indirect on gut via vit D). Vit D: ↑ both. Calcitonin: ↓ both. 99% of Ca is in bone. Of serum Ca, 50% is ionized and 40% is albumin-bound, so correct for albumin (+0.8 per 1 g/dL below 4). Low albumin = low total Ca without tetany.

🔗 **Link:** When PTH is secreted in excess, every arrow in that table is pushed too far.

👉 **Up next: Hyperparathyroidism.**

2.3 Hyperparathyroidism: Causes (Slides 9-10) ★

HPT = **excess PTH secretion.**

Type	Mechanism	Causes
1. Primary	Autonomous gland	★ Adenoma (90%) · hyperplasia · carcinoma · part of MEN I or II
2. Secondary	Appropriate response to a tendency to low serum Ca	★ Chronic renal failure ("renal osteodystrophy") · malabsorption and post-gastrectomy · [added] vit D deficiency
3. Tertiary	Autonomous PTH secretion superimposed on long-standing 2ry HPT	Usually CKD / post-transplant [added]
4. Ectopic PTH	Tumour secretes PTH	Very rare: bronchogenic carcinoma, hypernephroma (RCC)

Most malignancy-associated hypercalcaemia is due to PTHrP, which is not detected by the PTH assay, rather than true ectopic PTH. Slide 29 lists "ectopic PTH secretion (PTHrP)" in bronchogenic carcinoma.

[added] MEN 1 (Wermer) = Parathyroid, Pituitary, Pancreas. MEN 2A (Sipple) = Medullary thyroid, Pheochromocytoma, Parathyroid. MEN 2B has no parathyroid involvement.

Hyperparathyroidism: who started it, and what is the calcium?

PRIMARY

gland is the problem

Adenoma 90% · hyperplasia
carcinoma · MEN 1 / 2A

Ca ↑ · PO₄ ↓ · ALP ↑ · PTH ↑/=

Stones, bones, groans, moans
Osteitis fibrosa cystica

Rx: SURGERY (first line)

Screen for MEN

Unfit: cinacalcet + alendronate

Post-op: hungry bone syndrome

SECONDARY

normal gland responding to low Ca

CKD (renal osteodystrophy)
Malabsorption, post-gastrectomy
Vit D deficiency

Ca ↓/= · PO₄ ↑ (CKD) · ALP ↑ · PTH ↑

Bone pain, fractures

Rx: MEDICAL (treat cause)

PO₄ binder (sevelamer)

Ca supplement · vit D (calcitriol)

TERTIARY

secondary that went autonomous

Long-standing 2ry HPT (CKD)
e.g. persists after transplant *

Ca ↑ · PO₄ ↑ · ALP ↑ · PTH ↑↑

Hypercalcaemia despite CKD

Rx: SURGERY (first line)

Cinacalcet as adjunct

* added

Key discriminator = serum Ca: 1ry ↑ with PO₄ ↓ · 2ry ↓ or normal · 3ry ↑ with PO₄ ↑ (on a CKD background)

Ectopic PTH (bronchogenic carcinoma, hypernephroma) is very rare. Most malignancy secretes PTHrP, so true PTH is LOW.

Added diagram: primary vs secondary vs tertiary HPT

2.4 Clinical Picture (Slides 11-13) ★

Clinical pearl (slide 13): ★ **The majority of patients are asymptomatic** (found on routine Ca testing [added]).

Mnemonic: "Stones, bones, abdominal groans, thrones and psychiatric overtones"

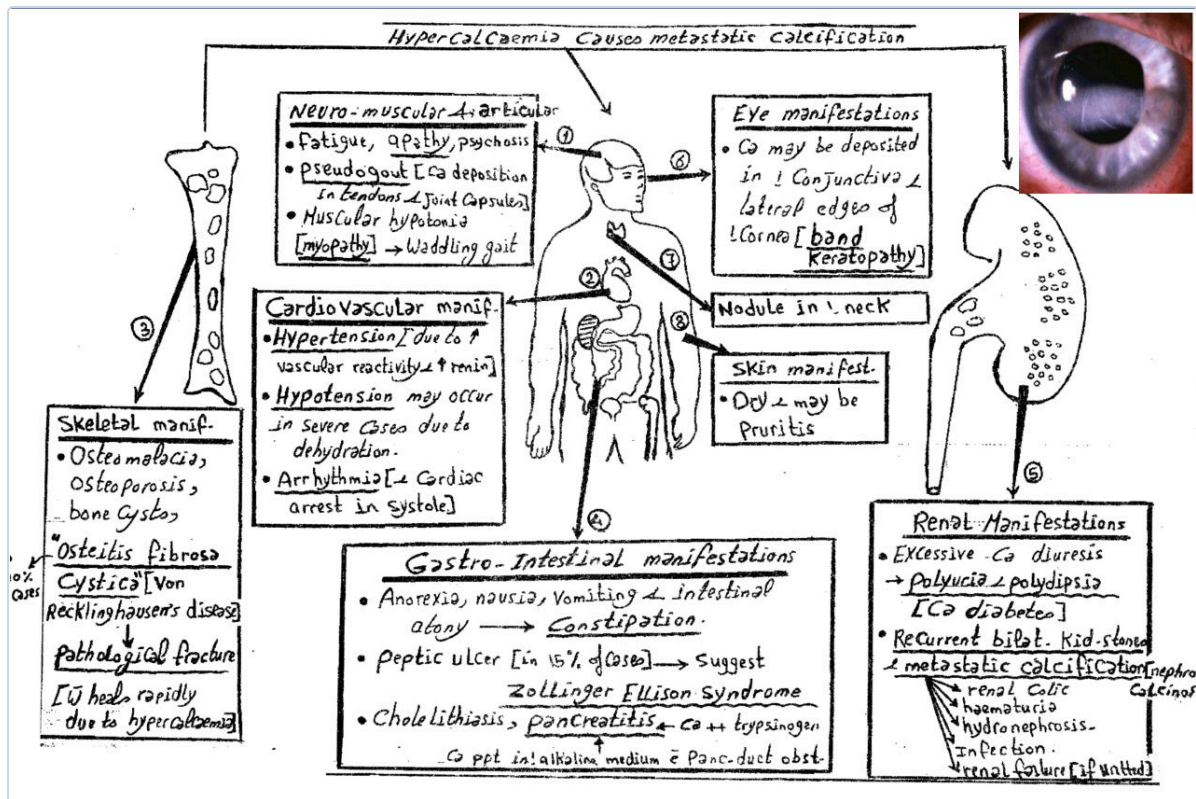
(slides 12–13)

- **Painful bones**
- **Renal stones**
- **Abdominal groans**
- **Sitting on the throne** (polyuria, constipation)
- **Psychiatric overtones**

Slide 11 (hand-drawn) and slide 12 by system. Hypercalcaemia causes **metastatic calcification**.

System	Manifestations
Skeletal	Osteomalacia, osteoporosis, bone cysts ; ★ osteitis fibrosa cystica (von Recklinghausen's disease of bone , ~10% of cases); pathological fractures , which heal rapidly because of the hypercalcaemia; bone pain
Renal	Excessive Ca diuresis → polyuria + polydipsia ("Ca diabetes"); ★ recurrent bilateral kidney stones and nephrocalcinosis → renal colic, haematuria, hydronephrosis, infection, renal failure if untreated
GIT	Anorexia, nausea, vomiting, intestinal atony → constipation ; ★ peptic ulcer (in 15% of cases → suggests Zollinger-Ellison/MEN 1); cholelithiasis ; pancreatitis (Ca activates trypsinogen; Ca precipitates in alkaline medium → duct obstruction)
Neuromuscular / articular	Fatigue, apathy , drowsiness, depression, psychosis ; pseudogout (Ca deposition in tendons and joint capsules); muscular hypotonia (myopathy) → waddling gait
Cardiovascular	Hypertension (↑ vascular reactivity, ↑ renin); hypotension in severe cases (dehydration); arrhythmia (cardiac arrest in systole); ECG: ★ short QT
Eyes	Ca deposits in the conjunctiva and lateral corneal edges → ★ band keratopathy
Neck	Nodule in the neck (rarely palpable; <i>[added]</i> a palpable mass with very high Ca suggests carcinoma)
Skin	Dry, pruritus (itching)

- PTH also **lowers blood phosphorus**.
- When **Ca × P rises**, calcium phosphate precipitates in tissues → **metastatic calcification**.



Slide 11: clinical picture of hyperparathyroidism (hand-drawn)

Clinical pearl

The majority of patients are asymptomatic.

Student pearl

"Stones, bones, abdominal groans, thrones, and psychiatric overtones!"

USMLE-style Question

Key learnings: signs of hypercalcaemia

- Painful bones
- Renal stones
- Abdominal groans
- Sitting on the throne (polyuria, constipation)
- & Psychiatric overtones.



Slide 13: clinical and student pearls; "stones, bones, groans, thrones, psychiatric overtones"

2.5 Osteitis Fibrosa Cystica (Slide 14) ★

- **Classic localised bone lesion of HPT.** PTH-driven **osteoclasts** lyse bone, and the **marrow is replaced by highly vascular fibrous tissue**. Stress on the weakened bone → **haemorrhage and cyst formation**.
- Old term: ★ "**brown tumour**". The brown colour comes from **massive haemosiderin deposition**.

- Typically in the **jaw and long bones**; may cause **pathological fractures**.
- Distinguished from other **giant cell tumours of bone** by measuring **serum Ca**.



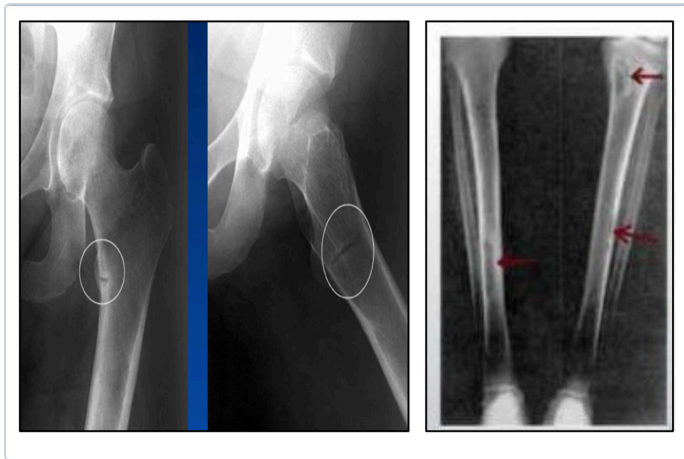
Slide 14: osteitis fibrosa cystica (brown tumours, tibia)

2.6 Investigations (Slides 15-20) ★

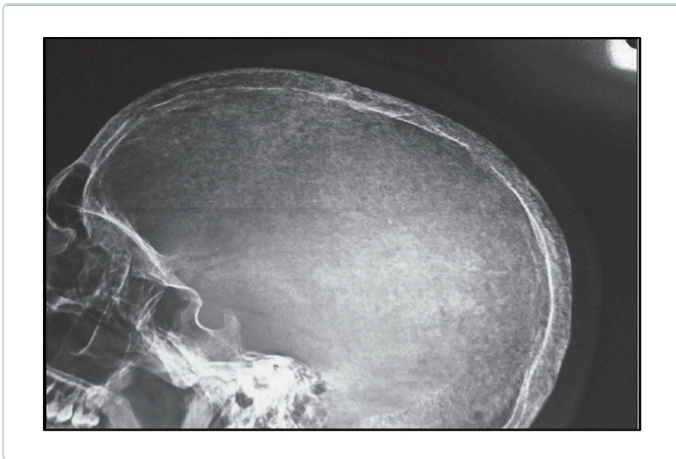
#	Test	Finding
1	Serum Ca	↑ (slide 15 gives N = 9–10.5 mg%)
2	Serum phosphate	↓ (except with renal failure). N = 2.5–4 mg%
3	Alkaline phosphatase	↑ in patients with bone disease
4	★ Plasma PTH	↑. Urinary cAMP ↑ in HPT but not in other causes of hypercalcaemia
5	Hypercalciuria	> 300 mg/24 h on a low-Ca diet is consistent with HPT (not diagnostic)
6	Radiology	See below
7	Localisation of the adenoma	Arteriography · US and CT of the neck · technetium-thallium subtraction scan
8	★ Steroid suppression (Dent) test	Hypercalcaemia of HPT does NOT suppress with hydrocortisone , unlike other hypercalcaemic states

Radiological findings (slide 16):

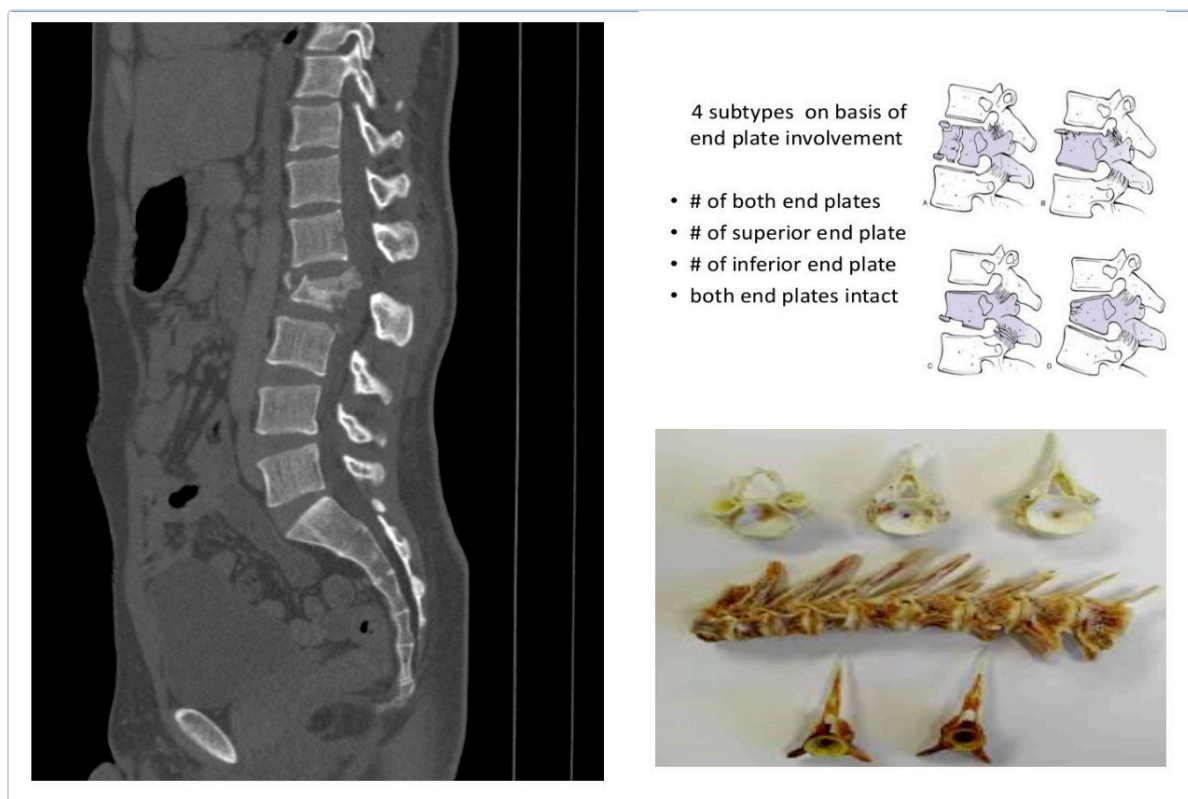
- **A. Kidneys: stones and nephrocalcinosis.**
- **B. Skeleton:** - **Osteitis fibrosa cystica (generalisata).** - ★ **"Salt and pepper" skull:** tiny punched-out lesions. - ★ **"Cod-fish spine":** diffuse softening; the discs indent the vertebral bodies. - **Milkman fracture / Looser zone (pseudofracture):** a tongue of radiolucency extending into the bone from the surface, mostly at the **upper femur and humerus** (due to osteomalacia). - [added] **Subperiosteal resorption of the radial side of the middle phalanges** is the most specific X-ray sign of HPT. Also distal clavicle resorption.
- **C. Soft tissue calcification.**



Slide 17: Looser zones (femoral necks) and long-bone lesions



Slide 18: salt-and-pepper skull

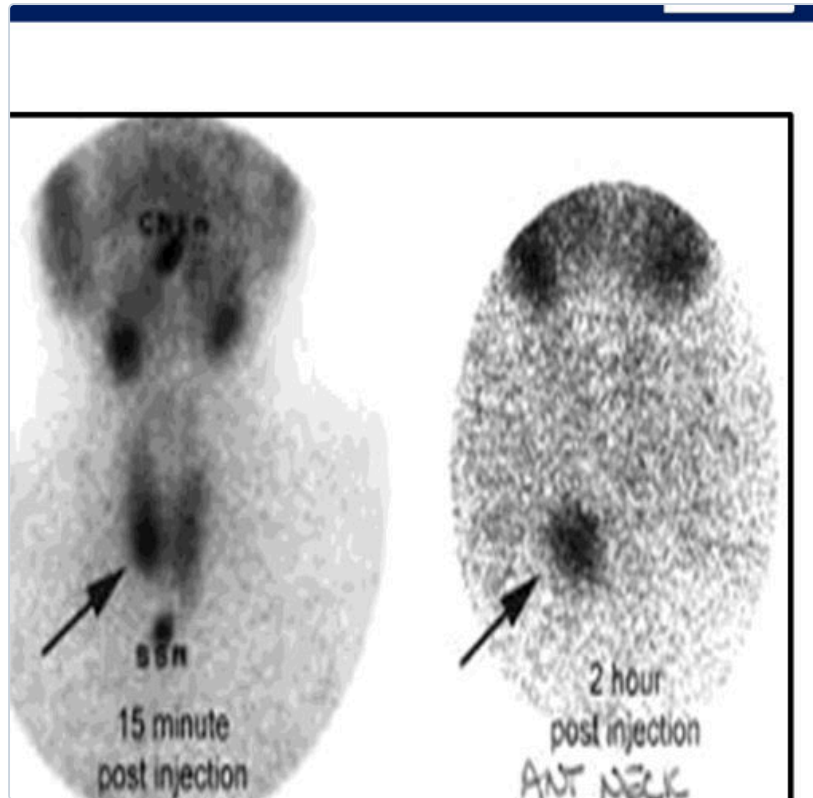


Slide 19: vertebral collapse on CT; end-plate fracture subtypes; cod-fish vertebrae

Slide 19: 4 subtypes of vertebral fracture by end-plate involvement: (1) both end plates · (2) superior end plate · (3) inferior end plate · (4) both end plates intact.

Localisation (slide 20):

- **Technetium visualises only the thyroid; thallium visualises thyroid AND parathyroid.** Subtracting one from the other leaves the parathyroid.
- [added] The modern standard is the ^{99m}Tc-sestamibi scan. The adenoma retains tracer on the 2-hour delayed image while the thyroid washes out, as shown on the slide's 15-minute vs 2-hour images.
- ★ **Clinical pearl: parathyroid imaging is for surgical planning, NOT for the initial diagnosis.** The diagnosis is **biochemical** (Ca + PTH).



Slide 20: parathyroid scintigraphy, 15 min vs 2 h

Recap: HPT diagnosis. 1ry = adenoma 90% (also hyperplasia, carcinoma, MEN). 2ry = CKD, malabsorption. 3ry = autonomous after long 2ry. Ectopic PTH is very rare. Most patients are asymptomatic; otherwise stones, bones, groans, moans, thrones. Brown tumours = osteitis fibrosa cystica (haemosiderin). Labs: $\uparrow$ Ca, $\downarrow$ PO₄, $\uparrow$ ALP, $\uparrow$ PTH, urinary cAMP $\uparrow$, hypercalciuria. X-ray: salt-and-pepper skull, cod-fish spine, stones. Imaging only localises for surgery. Dent test: HPT hypercalcaemia does not suppress with steroids.

Link: Confirmed 1ry HPT with indications is a surgical disease. Surgery brings its own early complication.

Up next: Treatment of HPT.

2.7 Treatment: Surgical (Slides 21-22) ★

Indications: all symptomatic patients, and asymptomatic patients with ANY of:

Criterion	Threshold
Age	< 50 years
Hypercalcaemia	Serum Ca > 1 mg/dL above the ULN
Renal involvement	eGFR < 60 mL/min · hypercalciuria · nephrolithiasis or nephrocalcinosis on imaging
Skeletal involvement	Reduced BMD (T-score ≤ -2.5 at any site) · vertebral fracture

Mnemonic: "50 - 1 - 60 - 2.5" = age < **50**, Ca **1** above ULN, GFR < **60**, T-score $\leq$ **-2.5**, plus any stone or fracture.

Operation (slide 22):

- **Adenoma → removed.**
- **Hyperplasia → subtotal resection:** 3 glands removed and the 4th partially removed (3½-gland parathyroidectomy).

★ **Complication: hungry bone syndrome (slide 22)**

- **Cause:** rapid **mineralisation of decalcified bone**, in patients with **bone disease before surgery**.
- **Presentation: tetany** due to **hypocalcaemia + hypomagnesaemia**.
- **Prevention:** give **vitamin D before the operation**.
- **Treatment of post-op tetany: IV Ca + MgSO₄** (2–4 mL of 50% solution IV infusion every 4 h).

[added] Also distinguish transient hypoparathyroidism from suppression of the remaining glands: PTH is low in that case, while in hungry bone PTH is normal or high and PO₄ is low.

2.8 Treatment: Medical (Slide 23, Summary Slide 27)

Indications:

- Previous **unsuccessful surgery**
- Patient **refuses** the operation
- Patient **unfit** for surgery

Drugs listed (slide 23):

- **Orthophosphate salts** 1–3 g/day
- **Oestrogen** in women (found to ↓ serum Ca in 1ry HPT)
- **Dichloroethylene diphosphate** (experimental)

Summary-slide (27) modern options: ★ **cinacalcet** (a **calcimimetic** that lowers Ca and PTH) + **alendronate** (bisphosphonate for bone).

Slide 23 lists older therapy. For exams, answer cinacalcet (± bisphosphonate) as the medical option, as on the summary slide.

2.9 Hypercalcaemic Crisis (Slides 24–26) ★

Definition: serum Ca > **13 mg%** (slide 24). The caution box on slide 25: **start immediate treatment if Ca > 14 mg/dL (> 3.5 mmol/L)**.

Causes:

1. **1ry HPT**
2. **Osteolytic carcinoma**
3. **Multiple myeloma**

Manifestations:

- **GIT:** abdominal pain, vomiting, constipation
- **Polyuria → dehydration**, weakness
- **Confusion → coma**
- **CVS:** arrhythmia, ★ **shortened QT**

Treatment (slides 25–26):

1. Agents that ↑ urinary Ca excretion

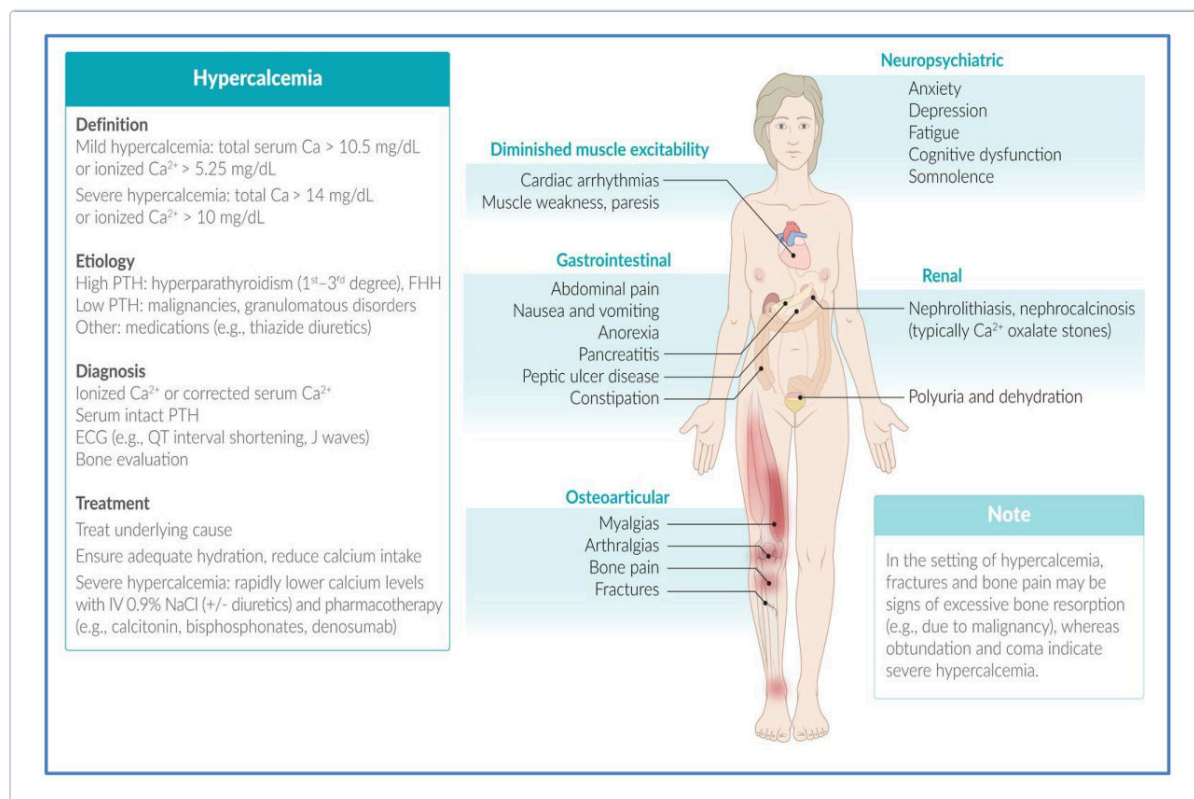
Step	Detail
a) ★ Hydration with isotonic saline: THE FIRST GOAL	4-6 L/day may be needed
b) ★ Furosemide IV 100 mg/h after the patient is rehydrated	NEVER without saline infusion , or dehydration will worsen
c) Dialysis	For hypercalcaemia complicated by renal failure

2. Agents that affect bone Ca fluxes

Drug	Mechanism / indication
Mithramycin	Inhibits bone resorption; used particularly in malignant hypercalcaemia
PG inhibitors (indomethacin, aspirin)	Hypercalcaemia from osteolytic metastases
★ Bisphosphonates: IV pamidronate	Drug of choice (malignancy) . Inhibit osteoclastic activity (<i>the slide adds "↑ phosphate level → ↓ Ca"</i>)
Calcitonin (SC)	↑ Ca excretion in urine, ↓ Ca release from bone. <i>[added]</i> Fastest onset (hours); tachyphylaxis after 48 h
★ Corticosteroids (prednisone)	↓ bone resorption + ↓ intestinal absorption. First line for hypercalcaemia from vitamin D excess: myeloma, lymphoma, sarcoidosis. NOT beneficial in hyperparathyroidism

[added] Others: denosumab (RANKL antibody; for bisphosphonate-refractory disease or renal failure); cinacalcet for parathyroid carcinoma.

Slide 25 is incomplete on the slide ("Diuretics:", "Steroids (Cortisone): is.", "→ ↓ serum Ca"). Slide 26 gives the complete content, which is used above.



Slide 34: hypercalcaemia overview (definition, etiology, diagnosis, treatment)

Slide 34 definitions:

- **Mild:** total Ca > **10.5 mg/dL** or ionized > **5.25**.

- **Severe:** total Ca > **14 mg/dL** or ionized > **10**.
- **Etiology:** - High PTH: HPT (1ry-3ry), **FHH**. - Low PTH: **malignancy, granulomatous disease**. - Other: **thiazides**.
- **ECG:** short QT, J (Osborn) waves.
- **Treatment:** severe → **IV 0.9% NaCl (± diuretics)** + calcitonin, bisphosphonates, denosumab.
- **Note:** fractures and bone pain suggest excessive bone resorption (e.g., malignancy); **obtundation and coma indicate severe hypercalcaemia**.

🔄 **Recap: Treatment and crisis.** Operate on symptomatic patients and on asymptomatic ones with age < 50, Ca > 1 above ULN, GFR < 60, stones/hypercalciuria, or T ≤ -2.5/fracture. Adenoma → excise; hyperplasia → 3½ glands. Hungry bone → tetany (low Ca + low Mg) → IV Ca + MgSO₄; prevent with pre-op vit D. Medical: cinacalcet ± alendronate. Crisis (> 13-14): saline first (4-6 L), furosemide only after rehydration, bisphosphonate (pamidronate) for malignancy, calcitonin, steroids for vit D-mediated causes (not HPT), dialysis if renal failure.

🔗 **Link:** HPT is only one cause of a high Ca. The differential is broad, and bone disease mimics it.

👉 **Up next: Differential diagnosis.**

2.10 Differential Diagnosis: Causes of Hypercalcaemia (Slides 28-31) ★

Group	Causes
1. Endocrine	Hyperparathyroidism · hyperthyroidism · Addison's disease · acromegaly
2. Malignant	★ Osteolytic metastases: local bone destruction via prostaglandins or osteoclast-activating factor (OAF) · ★ multiple myeloma (OAF) · some lymphomas secrete 1,25(OH)₂D₃ · ectopic PTH/PTHrP (e.g., bronchogenic carcinoma)
3. Iatrogenic	Hypervitaminosis D (treated with cortisone , which blocks vit D action on bone and intestine) · hypervitaminosis A · ★ milk-alkali (Burnett) syndrome · <i>[added]</i> thiazides, lithium
4. Miscellaneous	Sarcoidosis · Paget's disease of bone (only if immobilised) · immobilisation (paraplegia, fracture) · idiopathic infantile hypercalcaemia (IIH)

Milk-alkali (Burnett) syndrome (slide 30). Seen in peptic ulcer patients who take excess milk and alkali for a long time.

- **Early:** mild **alkalosis, hypercalcaemia, hypercalciuria**.
- **Later:** **nephrocalcinosis**, renal damage, **uraemia** → ↓ **serum Ca** and **renal acidosis**.
- *[added]* Modern version: **calcium carbonate supplements** for osteoporosis or dyspepsia.

Idiopathic infantile hypercalcaemia (slide 31):

Form	Cause	Picture
Reversible	↑ vit D intake by the mother during lactation	Failure to thrive, vomiting, abdominal pain, constipation
Irreversible	↑ vit D intake by the mother during pregnancy	★ Supravalvular aortic stenosis, elfin facies , mental retardation → renal failure . <i>[added]</i> This is the Williams syndrome phenotype (7q11.23 deletion); the "maternal vit D" mechanism on the slide is historical

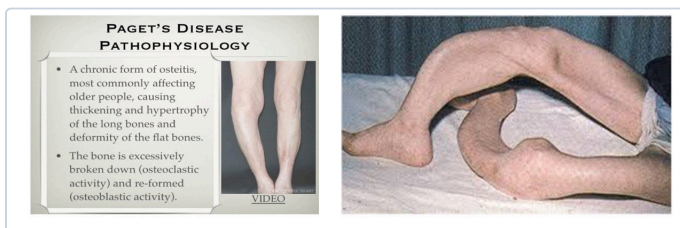
Mnemonic for hypercalcaemia: "CHIMPANZEES" [added]

- **C**alcium supplements (milk-alkali)
- **H**yperparathyroidism / **H**yperthyroidism
- **I**atrogenic (thiazides) / **I**mmobilisation
- **M**ultiple myeloma / **M**ilk-alkali
- **P**aget's (immobilised)
- **A**ddison's / **A**cromegaly
- **N**eoplasm
- **Z**ollinger-Ellison (MEN 1)
- **E**xcess vit **D**
- **E**xcess vit **A**
- **S**arcoidosis

2.11 Causes of Generalised Bone Disease (Slides 32-33)

These conditions can mimic HPT bone disease:

1. **Renal osteodystrophy** (see chronic renal failure)
2. **Osteomalacia, osteoporosis**
3. **Paget's disease of bone**: - **High ALP** - **↑ Ca only if the patient is immobilised** - X-ray: **lytic areas alongside dense areas** (new bone formation) - Slide figure: chronic osteitis of older people, with thickening, hypertrophy and **deformity of long and flat bones**, e.g. bowed tibia
4. **McCune-Albright syndrome** (rare): - **Fibrous dysplasia (unilateral)** - **Areas of pigmentation** ([added] café-au-lait with a "coast of Maine" border) - **Precocious puberty (in girls)**
5. **Osteoblastoma**: radiological appearance **similar to osteitis fibrosa cystica but with NO biochemical changes**



Slide 32: Paget's disease of bone



Slide 33: McCune-Albright fibrous dysplasia of the hand

2.12 Case & Summary (Slides 27, 35-38)

Case (slides 35-37):

- **Patient**: 63-year-old woman on **dialysis** with **CKD**, DM, HTN and depression, on **sevelamer** and **sodium bicarbonate**.

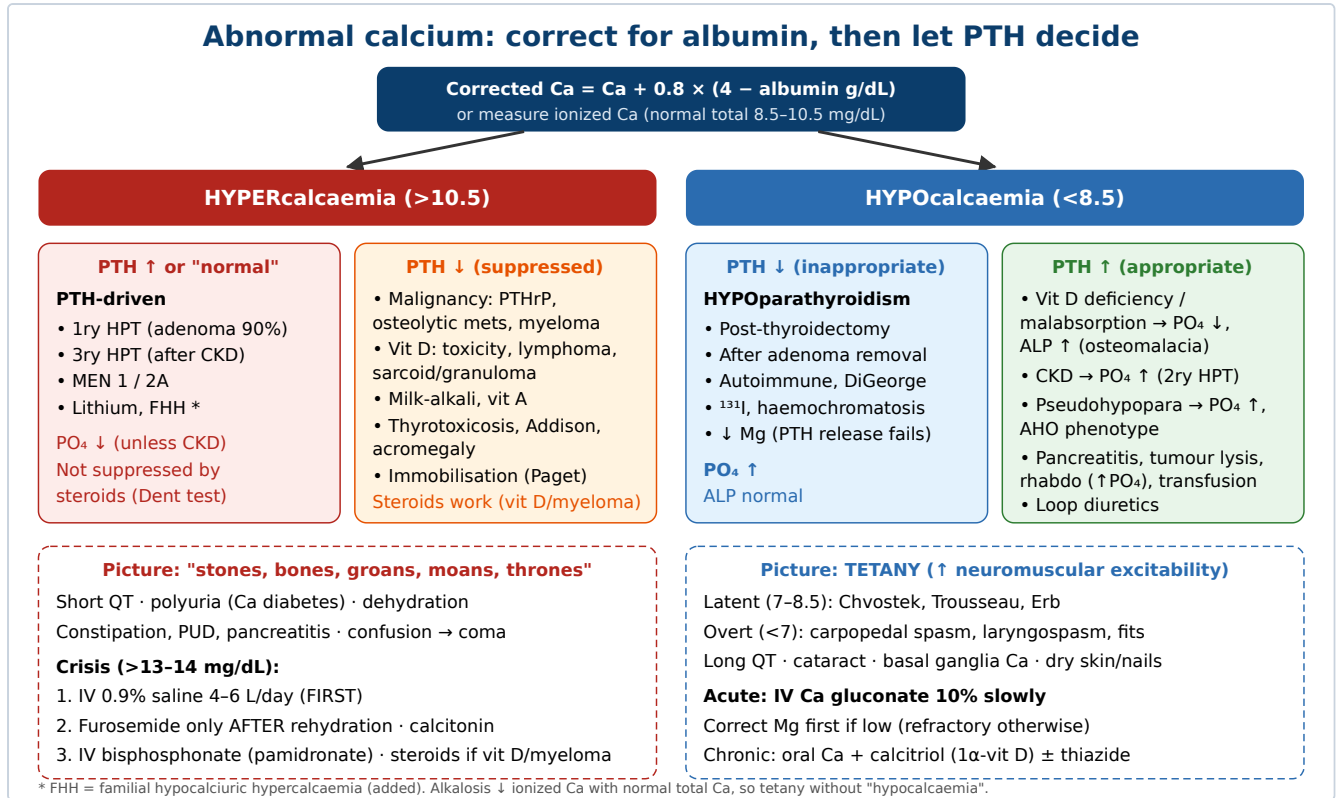
- **Presentation:** severe low back pain, L2 tenderness. X-ray: **wedge compression fracture of L2.**
- **Labs:** Ca **8.2**, creatinine **3.8**, BUN 33, Hb 10.1, platelets 130 000.
- **Question:** most likely explanation? (*Answered in §7, Q1.*)

Summary table (slide 27) ★

Type	Ca	PO ₄	ALP	PTH	Management
Primary	↑	↓	↑	↑ / =	Screen for MEN; surgical removal is first line; unfit → alendronate + cinacalcet
Secondary	↓	↑	↑	↑	Medical therapy first line , treat the cause. CKD: phosphate binders (sevelamer), Ca supplements, vit D replacement
Tertiary	↑	↑	↑	↑	Surgical removal first line; cinacalcet as adjunct

Type	Ca	PO ₄	ALP	PTH	Management
Primary hyperparathyroidism	↑	↓	↑	↑/=	• Screen for MEN • Surgical removal of parathyroid glands is first line • If patients are unfit for surgery, medical therapy with alendronate and cinacalcet (calcimimetic agent that lowers calcium and PTH) may be attempted
Secondary hyperparathyroidism	↓	↑	↑	↑	• Medical therapy is first line – treat underlying cause • Treatment with CKD: 1. Phosphate binders such as sevelamer 2. Calcium supplementation 3. Vit D replacement
Tertiary hyperparathyroidism	↑	↑	↑	↑	• Surgical removal of the gland is first line • Cinacalcet may be used as adjunctive therapy

Slide 27: summary of 1ry / 2ry / 3ry HPT



Added diagram: PTH-first approach to abnormal calcium

Recap: Differential.

- **Endocrine:** HPT, thyrotoxicosis, Addison's, acromegaly.
- **Malignancy:** osteolytic mets (PG/OAF), myeloma (OAF), lymphoma (calcitriol), PTHrP.
- **Iatrogenic:** vit D/A, milk-alkali.
- **Miscellaneous:** sarcoid, immobilisation, Paget's if immobilised, IIH.
- **Bone mimics:** renal osteodystrophy, osteomalacia, osteoporosis, Paget's (↑ ALP), McCune–Albright, osteoblastoma (normal biochemistry).
- **The three HPT types:** 1ry ↑Ca ↓PO₄ → surgery; 2ry ↓Ca ↑PO₄ → binders + Ca + vit D; 3ry ↑Ca ↑PO₄ → surgery.

3. High-Yield Comparison Tables

Table A: Lab Patterns

Condition	Ca	PO ₄	ALP	PTH	Extra
1ry HPT	↑	↓	↑ (if bone disease)	↑ / inappropriately normal	↑ urinary cAMP, hypercalciuria
2ry HPT (CKD)	↓ / N	↑	↑	↑	↓ calcitriol, ↑ creatinine
2ry HPT (vit D def / malabsorption)	↓ / N	↓	↑	↑	↓ 25-OH-D [added]
3ry HPT	↑	↑	↑	↑↑	CKD / post-transplant
Malignancy (PTHrP)	↑↑	↓	±	↓	↑ PTHrP [added]

Condition	Ca	PO ₄	ALP	PTH	Extra
Osteolytic mets / myeloma	↑	N / ↑	↑ (mets) / N (myeloma) [added]	↓	M-protein, Bence Jones
Vit D excess / granuloma / lymphoma	↑	↑	N	↓	↑ 1,25-D; steroid-responsive
Milk-alkali	↑	N / ↑	N	↓	Metabolic alkalosis , renal impairment
Paget's	N (↑ if immobilised)	N	↑↑	N	Lytic + sclerotic X-ray
Osteoblastoma	N	N	N	N	X-ray mimics OFC

Table B: Hypercalcaemia Drugs

Drug	Onset [added]	Best for	Caution
IV normal saline	Immediate	ALL: first step	Fluid overload in HF/CKD
Furosemide	Hours	After rehydration	Never before saline
Calcitonin (SC)	4-6 h	Bridge while bisphosphonate acts	Tachyphylaxis
IV pamidronate / zoledronate	2-4 days	★ Malignancy (drug of choice)	Renal function, jaw ONJ [added]
Glucocorticoids	Days	★ Vit D excess, myeloma, lymphoma, sarcoid	Useless in HPT
Mithramycin		Malignant (historical)	Toxic
Indomethacin/aspirin		Osteolytic mets (PG-mediated)	
Dialysis	Immediate	Renal failure	
Cinacalcet		1ry HPT unfit for surgery, 3ry, parathyroid Ca	Hypocalcaemia

Table C: Primary vs Secondary vs Tertiary

	Primary	Secondary	Tertiary
Driver	Gland (adenoma 90%)	Low Ca (CKD, malabsorption)	Autonomous after long 2ry
Ca	↑	↓ / N	↑
PO ₄	↓	↑ (CKD)	↑
First-line Rx	Surgery	Medical	Surgery
Medical option	Cinacalcet + alendronate	Sevelamer + Ca + vit D	Cinacalcet (adjunct)

Table D: Radiology Signs

Sign	Meaning
Salt-and-pepper skull	HPT (punched-out lesions)
Brown tumour (osteitis fibrosa cystica)	HPT; haemosiderin; jaw and long bones
Cod-fish spine	Softened vertebrae indented by discs (HPT / osteomalacia)

Sign	Meaning
Looser zone / Milkman fracture	Pseudofracture of osteomalacia (upper femur, humerus)
Nephrocalcinosis / stones	Hypercalciuria
<i>[added]</i> Rugger-jersey spine	Renal osteodystrophy (2ry HPT)
<i>[added]</i> Subperiosteal resorption, radial middle phalanges	Most specific sign of HPT
Lytic + dense areas, ↑ ALP	Paget's

4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Low total Ca → tetany	Low albumin (cirrhosis, nephrotic) → low total, normal ionized → no tetany
PTH lowers PO ₄	In renal failure PO ₄ stays high (2ry / 3ry HPT)
HPT → ↑ Ca	2ry HPT → Ca low or normal
HPT patients are symptomatic	★ Most are asymptomatic
Fractures heal slowly in bone disease	HPT fractures heal rapidly (hypercalcaemia) (slide 11)
Steroids lower Ca	Not in HPT (basis of the Dent test); they work for vit D-mediated causes (myeloma, lymphoma, sarcoid, vit D toxicity)
Diuretics treat hypercalcaemia	Loop only, after saline. Thiazides raise Ca
Paget's has normal Ca	↑ Ca if immobilised
Imaging finds the adenoma	Use only for surgical planning , never for diagnosis
Parathyroid surgery cures	Watch for hungry bone → hypocalcaemia + hypomagnesaemia
Malignancy hypercalcaemia = ectopic PTH	Usually PTHrP ; true ectopic PTH is very rare
Osteoblastoma looks like OFC	Biochemistry is normal

5. Clinical Correlations

- Routine bloods show Ca 11.2, low PO₄, high PTH in an asymptomatic 45-year-old** → 1ry HPT. Age < 50 → **parathyroidectomy**. Image (sestamibi/US) only to plan the operation.
- Recurrent renal stones + peptic ulcers + pituitary adenoma** → **MEN 1**.
- HPT + medullary thyroid cancer + pheochromocytoma** → **MEN 2A**. *[added]* **Remove the pheo first.**
- Dialysis patient with bone pain, high PO₄, high PTH, low/normal Ca** → **2ry HPT / renal osteodystrophy** → sevelamer, calcitriol, Ca, cinacalcet.
- Renal transplant recipient whose Ca is now high** → **3ry HPT** → parathyroidectomy.
- Smoker with hilar mass, Ca 14, low PTH** → **squamous cell carcinoma** → **PTHrP** → saline + bisphosphonate. *[Lung cancer lecture link]*
- Elderly patient with back pain, anaemia, renal impairment, hypercalcaemia** → **myeloma** → saline + bisphosphonate + **steroids**.
- Sarcoidosis patient with hypercalcaemia** → granuloma 1α-hydroxylase → **steroids**.
- Patient taking antacids + milk for an ulcer, with alkalosis, high Ca and renal impairment** → **milk-alkali**.

- **Day 2 after parathyroidectomy: tingling, Chvostek positive → hungry bone → IV Ca gluconate + MgSO₄.**
- **Confused, vomiting, polyuric patient with short QT and Ca 15 → crisis → IV saline first.**

6. Rapid Revision (Last-Day)

1. **PTH:** ↑Ca ↓PO₄ (direct bone + kidney; indirect gut via vit D). **Vit D:** ↑ both. **Calcitonin:** ↓ both.
2. 99% of Ca is in bone. Serum Ca: **50% ionized**, 40% albumin-bound, 10% complexed.
3. **Corrected Ca = Ca + 0.8 × (4 – albumin).** Low albumin → no tetany.
4. Normal total Ca **8.5-10.5 mg/dL**; ionized **4.65-5.25 mg/dL**; PO₄ **3-4.5**.
5. **1ry HPT:** adenoma **90%**, hyperplasia, carcinoma, MEN 1/2A. **2ry:** CKD, malabsorption, gastrectomy. **3ry:** autonomous after 2ry. **Ectopic:** bronchogenic carcinoma, RCC (rare).
6. **Most patients are asymptomatic.** Otherwise **stones, bones, abdominal groans, thrones, psychiatric overtones.**
7. **Ca diabetes** (polyuria), **band keratopathy**, **pseudogout**, **short QT**, HTN, **PUD (ZES)**, **pancreatitis**, constipation, dry itchy skin.
8. **Osteitis fibrosa cystica = brown tumour** (haemosiderin; jaw and long bones).
9. Labs: ↑Ca, ↓PO₄ (unless CKD), ↑ALP (bone), ↑PTH, **↑ urinary cAMP**, urine Ca **> 300 mg/24 h**.
10. X-ray: **salt-and-pepper skull**, **cod-fish spine**, stones, nephrocalcinosis, soft-tissue Ca. **Looser zones = osteomalacia.**
11. Localisation: US/CT neck, **Tc-thallium** (subtraction), arteriography. **Imaging is for surgery, not diagnosis.**
12. **Dent test:** HPT hypercalcaemia is **not steroid-suppressible.**
13. **Surgery if symptomatic**, or asymptomatic with **age < 50, Ca > 1 above ULN, eGFR < 60, hypercalciuria/stones, T ≤ -2.5, vertebral fracture.**
14. Adenoma → excise; hyperplasia → **3½ glands.**
15. **Hungry bone:** tetany (↓Ca + ↓Mg) → **IV Ca + MgSO₄**; prevent with **pre-op vit D.**
16. Medical: **cinacalcet + alendronate** (older: orthophosphate, oestrogen).
17. **Crisis > 13 (treat immediately > 14): saline 4-6 L FIRST** → furosemide after rehydration → **pamidronate (DOC in malignancy)**, calcitonin, **steroids for vit D excess/myeloma/lymphoma/sarcoid (not HPT)**, dialysis if renal failure.
18. Hypercalcaemia causes: endocrine (HPT, thyrotoxicosis, Addison, acromegaly), malignant, iatrogenic (vit D/A, **milk-alkali**), miscellaneous (sarcoid, immobilisation, Paget's if immobilised, IIH).
19. Bone mimics: renal osteodystrophy, osteomalacia, osteoporosis, **Paget's (↑ALP)**, **McCune-Albright** (fibrous dysplasia + pigmentation + precocious puberty), **osteoblastoma (normal biochemistry).**
20. **1ry ↑Ca ↓PO₄ → surgery · 2ry ↓Ca ↑PO₄ → medical · 3ry ↑Ca ↑PO₄ → surgery.**

7. Exam-Style Questions

Q1 (lecture case, slides 35-37). A 63-year-old woman on dialysis for CKD (taking sevelamer and sodium bicarbonate) has an L2 wedge compression fracture. Ca is 8.2 mg/dL and creatinine 3.8 mg/dL. What is the most likely explanation? A. Tertiary HPT · B. Senile osteoporosis · C. Secondary HPT · D. Postmenopausal osteoporosis · E. Elder abuse · F. Pseudohypoparathyroidism · G. Primary HPT

Answer: C. Secondary hyperparathyroidism (renal osteodystrophy). *[reasoning added; key not on slides]*

- In CKD, ↓ calcitriol and ↑ PO₄ lead to low/normal Ca and a chronic rise in PTH. That causes high-turnover bone loss and fragility fractures.
- She is on **sevelamer**, a phosphate binder for 2ry HPT (slide 27).
- **Tertiary** and **primary** HPT would give **high** Ca; hers is 8.2.
- Osteoporosis alone ignores the CKD-mineral bone disorder.
- The family conflict is a distractor for **elder abuse**; fracture pattern and context fit metabolic bone disease.
- **Pseudohypoparathyroidism** is congenital, with AHO features.

Q2. A 52-year-old woman has kidney stones. Ca 11.6 mg/dL (ULN 10.5), PO₄ 2.1, PTH raised, eGFR 85, DEXA T-score -1.8. Which is the most appropriate management? A. Observation with yearly Ca · B. Parathyroidectomy · C. Cinacalcet · D. Prednisolone · E. Sestamibi scan to confirm the diagnosis

Answer: B.

- This is **1ry HPT** with **nephrolithiasis**, and her Ca is **> 1 mg/dL above the ULN**. Either alone is a surgical indication (slide 21).
- **Cinacalcet** is for patients unfit for or refusing surgery.
- **Steroids don't work** in HPT.
- **Imaging is for localisation, not diagnosis** (slide 20).

Q3. A 70-year-old man is confused and vomiting. Ca 15.2 mg/dL, dry mucosa, BP 98/60, creatinine mildly raised. What is the FIRST step? A. IV furosemide · B. IV pamidronate · C. IV 0.9% saline · D. SC calcitonin · E. Oral prednisolone

Answer: C.

- **Rehydration with isotonic saline is the first goal** (slide 26).
- **Furosemide must never precede saline**, because it worsens dehydration.
- Bisphosphonate and calcitonin come next.

Q4. Hypercalcaemia that does NOT fall after a course of hydrocortisone most likely comes from: A. Sarcoidosis · B. Vitamin D intoxication · C. Multiple myeloma · D. **Primary hyperparathyroidism** · E. Lymphoma

Answer: D.

- This is the **Dent steroid suppression test** (slide 20): **HPT hypercalcaemia does not suppress**.
- Steroids work in vit D-mediated causes (A, B, C, E) (slide 26).

Q5. The day after removal of a parathyroid adenoma, a patient with long-standing osteitis fibrosa cystica develops perioral tingling and carpopedal spasm. Ca is 6.9, Mg is low, PO₄ is low. What is the best treatment, and how could it have been prevented? A. Oral Ca; nothing · B. **IV Ca + MgSO₄; pre-operative vitamin D** · C. IV saline; pre-op bisphosphonate · D. Thiazide; pre-op cinacalcet · E. PTH injection; pre-op oestrogen

Answer: B.

- This is **hungry bone syndrome**: rapid remineralisation causes **hypocalcaemia + hypomagnesaemia**.
- Treat with **IV Ca + MgSO₄**; prevent with **vit D before surgery** (slide 22).

8. Exam Pitfalls & Traps

- **Normal-range conflicts in the lecture**: - Total Ca is given as **8.4-10.2** (slide 7), **8.5-10.5** (slides 7, 8, 34) and **9-10.5** (slide 15). Use **8.5-10.5 mg/dL** unless the question gives a range. - Ionized Ca: slide 8 says "< 4.5 mg/dL" with a normal of **4.65-5.25**.
- **Albumin units**: slide 7 writes "below 4 **mg/dL**". Albumin is in **g/dL**; the formula is unchanged.
- **Crisis threshold**: slide 24 says **> 13 mg%**; slide 25 and slide 34 say **> 14 mg/dL**. Know both and quote the source the question implies.
- **Slide 25 is incomplete** ("Diuretics:", "Steroids (Cortisone): is."). Slide 26 gives the full regimen.
- **Medical therapy**: slide 23 lists old drugs (orthophosphate, oestrogen, dichloroethylene diphosphate), while the summary slide uses **cinacalcet + alendronate**. Modern MCQs expect **cinacalcet**.
- **Bisphosphonate mechanism**: the slide says " $\uparrow$ phosphate level $\rightarrow$ $\downarrow$ Ca". The core mechanism is **osteoclast inhibition**.
- **Furosemide before saline** is a classic wrong answer.
- **Steroids for HPT hypercalcaemia** is a wrong answer. Steroids are right for **myeloma, lymphoma, sarcoid and vit D toxicity**.
- **2ry HPT has LOW/normal Ca**. A dialysis patient with **high** Ca $\rightarrow$ **tertiary**.
- **Thiazide-induced hypercalcaemia** can unmask 1ry HPT. Loop diuretics lower Ca.
- **Looser zones** belong to **osteomalacia**, even though slide 16 lists them under HPT radiology.
- **Paget's** gives very high ALP with **normal Ca** unless the patient is immobilised.
- **Osteoblastoma vs OFC**: same X-ray, but osteoblastoma has **normal labs**.
- **Irreversible IIH** (supraclavicular AS, elfin facies) = **Williams syndrome** phenotype. The slide's maternal-vit D explanation is outdated *[added]*.
- **Ectopic PTH** is "very rare". Most cancer hypercalcaemia = **PTHrP** (squamous lung, breast, RCC) or osteolysis, with **suppressed PTH**.